

Internship Report

Title: Quality Inspection and Defect Detection Using Computer Vision

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Organization: Godrej & Boyce Mfg. Co. Ltd.

Acknowledgment

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Abstract

This report presents the work carried out during my internship on the project titled 'Quality Inspection and Defect Detection Using Computer Vision'. The objective was to develop and deploy an AI-based tool capable of identifying various defects in aerospace parts, including scratches, dents, cracks, and surface anomalies. The project also involved implementing OCR validation using a vision-language model. The final application was deployed as a web-based tool accessible via Streamlit.

Introduction

In the aerospace industry, where safety and precision are non-negotiable, quality inspection serves as the foundation of every manufacturing process. Traditional manual inspections, however, can be laborious and inconsistent, placing a lot of strain and mental pressure on technicians and risking undetected defects.

Quality inspection is a critical process in manufacturing, especially in industries like aerospace where precision and safety are paramount. This project explores the use of computer vision and AI to automate the detection of defects in aerospace tools, thereby improving accuracy and efficiency.

Project Overview

The project aimed to build a computer vision system for detecting defects such as scratches, foreign object debris (FODs), surface defects, dents, and cracks in aerospace tools. Additionally, an OCR validation tool was developed to compare text extracted from images against a master reference using a vision-language model.

Tools and Technologies

The following tools and technologies were used during the internship:

- Python: A versatile, high-level programming language widely used for scripting, data analysis, and rapid prototyping.
- OpenCV: An open-source computer vision library that provides tools for real-time image and video processing.
- Streamlit: A framework for building and sharing interactive web applications in Python.
- Moondream: A specialized AI-driven imaging platform advanced visual data annotation and model training.
- NumPy: A foundational library for numerical computing in Python, offering efficient array objects and mathematical operations.
- Pillow: A fork of the Python Imaging Library (PIL) for opening, manipulating, and saving many different image file formats.
- Scikit-Image: A collection of algorithms for image processing in Python, built on top of NumPy and SciPy.

Methodology

The defect detection system was implemented using image processing techniques such as edge detection, blob detection, and texture analysis. Each type of defect was identified using an algorithm:

- Scratches: Detected using Canny edge detection and morphological operations.
- Dents and Cracks: Identified using Laplacian of Gaussian blob detection.
- Surface Defects: Detected using Local Binary Patterns (LBP) and thresholding.
- OCR Validation: Implemented using Moondream AI to extract and compare text from the master image wrt the sample images.

Implementation

The application was developed using Streamlit to provide an interactive web interface. Users can upload or capture images of aerospace tools and select the type of defect detection they want to perform. The backend processes the image and displays the metrics.

The OCR validation tool allows users to upload a test image and a master image. The Moondream model extracts text from both images and compares them to determine if they match.

The final application was deployed and is accessible at:

<https://defect-detection.streamlit.app>

Results and Discussion

The deployed application successfully detects various types of defects and provides a summary of the findings. The OCR validation tool accurately compares text from images and highlights mismatches. The use of Moondream AI significantly improved the accuracy of text extraction and validation. It also lets the user export the results of the defects if any.

Output Examples

OCR Validation Results

- **Status:** Pass/Fail with clear indicators
- **Match Percentage:** Numerical similarity score
- **Detailed Comparison:** Character-level differences
- **Extracted Text:** Side-by-side text comparison

Defect Detection Results

- **Summary:** Total defect count and types
- **Visual Overlays:** Defects highlighted on original image
- **Quality Metrics:** Brightness, contrast, sharpness, noise
- **Exportable Reports:** CSV format for further analysis

Challenges Faced

The deployment using Streamlit was tough due to there being a lot of parameters which needed to be fulfilled for it to be properly deployed. However, with proper configuration and testing, the deployment was completed successfully.

In addition to deployment hurdles, the OCR component proved particularly stubborn. My initial implementation used Tesseract OCR, but its accuracy on specialized aerospace images fell well below requirements. So, we then experimented with EasyOCR and PaddleOCR—both of which delivered marginal improvements yet still failed to reliably capture fine surface details and engraved markings. I ultimately found the Moondream AI model met the precision needs. Integrating Moondream required learning its API conventions and adjusting our image pipelines, but once in place it provided the consistency and accuracy essential for automated defect detection.

Learnings & Takeaways

During this internship, I gained hands-on experience in computer vision, image processing, and deploying AI models. I also learned how to build interactive web applications using Streamlit and how to integrate external APIs for advanced functionalities.

Future Scope

Future improvements could focus on enhancing the accuracy of defect detection using deep learning models. Additional features such as real-time video analysis and integration with manufacturing systems could also be explored.

- Machine Learning Models: Integration with custom trained models
- Batch Processing: Process multiple images simultaneously
- Real-time Detection: Live video stream analysis
- Advanced Analytics: Statistical analysis and reporting
- Cloud Integration: Remote processing capabilities

References

- OpenCV Documentation
- Streamlit Documentation
- Moondream AI Documentation
- Scikit-image Documentation